

2023



Aillis Achieves 10x Faster Inference Using PyTorch on AWS and Amazon EC2

Learn how Aillis built an AI device to detect influenza using NVIDIA GPU-based Amazon EC2 instances and PyTorch on AWS.

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80% reduction

in initial deployment cost for customers

Scaled

to run 50 AI models in parallel

10x faster

inference speed than with edge device



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Overview

[Aillis Inc.](#) (Aillis), a medical company based in Japan, set out in 2018 to build an artificial intelligence device to detect influenza infection. The device—which has been approved in Japan and is currently in clinical trials in the United States—uses pharyngeal images and clinical information to determine whether a patient has influenza.



company wanted to train a set of AI models that could run in parallel to provide a rapid diagnosis, which required a solution that was both reliable and scalable.

To meet its needs, Aillis chose to use Amazon Web Services (AWS) and successfully built a medical camera. The device, which replicates the pharynx examination and diagnostic techniques of skilled doctors, was released in December 2022 in Japan. Using AWS services powered by [NVIDIA](#), an [AWS Partner](#), Aillis runs 50 AI models in parallel, achieves inference speeds that are 10 times faster than those of edge devices, and detects influenza in a matter of seconds (depending on local internet speeds).



Opportunity | Using PyTorch on AWS to Build an AI Medical Device

Aillis was founded in 2017 and develops AI medical devices that replicate the diagnostic techniques of skilled doctors. The company's goal is to improve the accuracy of future diagnoses and contribute to improving medical care for all.

According to the 2016 [Global Burden of Disease](#) study, influenza was estimated to be responsible for more than 300,000 seasonal respiratory deaths per year. Timely and accurate diagnosis of the infection can prevent transmission and reduce mortality risk. Wataru Takahashi, senior AI engineer at Aillis, lost his son to a severe case of influenza infection because of medical limitations and decided to develop a new system for influenza detection.

Aillis chose to use PyTorch as its AI framework so that it could access cutting-edge AI technology quickly, and the company chose to use AWS for its proactive technical support plus simple-to-use and infrastructure. To build the device, Aillis used [PyTorch on AWS](#), an open-source deep learning



accelerates the process from machine learning research to model deployment. The AI camera detects symptoms of infectious disease instead of relying on pathogen levels. “Medical devices require a very high level of reliability, and using PyTorch on AWS, we could meet this requirement,” says Atsushi Fukuda, chief technology officer at Aillis.

Solution | Reducing Deployment Costs by 80% Using NVIDIA GPU-Based Amazon EC2 Instances

To train its AI model, Aillis collected data from more than 10,000 patients across 100 sites in clinical studies. For this training, Aillis used [Amazon Elastic Compute Cloud](#) (Amazon EC2), which provides secure and resizable compute capacity for virtually any workload. Specifically, the company used [Amazon EC2 P3 Instances](#), which accelerate machine learning and offer high-performance computing applications with NVIDIA V100 Tensor Core GPUs. Using Amazon EC2 P3 Instances, the company’s AI engineers could run trial-and-error training sessions from nearly anywhere at any time, including while traveling on bullet trains. Aillis found this solution more reliable than other GPU servers. To scale AI inference workloads, the company uses [Amazon EC2 G4 Instances](#) (powered by NVIDIA T4 GPUs), which are well suited for machine learning inference and graphics-intensive applications. The Amazon EC2 G4dn Instance is secure at the medical device level, another benefit for Aillis.

Aillis launched a cloud server for viewing pharyngeal images and inputting clinical information using [AWS Fargate](#), a serverless, pay-as-you-go compute engine. “Our service takes time to run inferences, and we have to provide scalability, so we decided to use AWS Fargate,” says Fukuda. Additionally, because of the automatic scaling feature of AWS Fargate, Aillis engineers can deploy code quickly without worrying about high-load failures. Aillis also built a shipping inspection management cloud system on AWS Fargate to remotely control the quality of its manufacturing partner. Aillis uses AWS Fargate to run its web API server and, running on Amazon EC2, uses [Amazon Elastic Container Service](#) (Amazon ECS)—a fully managed container orchestration service that simplifies deployment, management, and scaling of containerized applications—to run inference servers and virtually assure scalability.

Aillis collected more than 500,000 pharyngeal images from the more than 10,000 patients at 100 sites in clinical studies, and these images are stored on [Amazon Simple Storage Service](#) (Amazon S3), an object storage service built to retrieve virtually any amount of data from anywhere. Using Amazon S3, the company meets the data standards for medical software development and medical electronic devices as well as good data governance practices.



standards. “Using Amazon S3, we strictly control access to both training and testing data, which we have to prove to the government regulation team,” says Takahashi. Thus, Aillis achieves security and scalability using AWS.

Using PyTorch on AWS and NVIDIA GPU-based Amazon EC2 instances, Aillis achieved inference times that were up to 10 times faster than using edge devices. “In terms of scalability and reliability, we need to nearly guarantee the time it takes to get an answer from the AI so that doctors and patients are not waiting for results,” says Fukuda. By using AWS instead of AI edge devices, the company also lowered the initial deployment cost of the device for customers by 80 percent. Aillis achieved this performance using the deep technical support of AWS.

The biggest benefit for Aillis is the success of its AI device as a solution. In clinical studies, the AI model achieved 76 percent sensitivity and 88 percent specificity, outperforming three physicians. “It is possible that sensitivity early in the onset of the disease is at the same level 12 hours after onset,” says Takahashi. “Administering anti-influenza medications at the early-onset phase can help patients recover quickly and prevent severe illness.” The examination is also quick, producing results within seconds depending on the local internet capabilities. The AI camera is an approved medical device in Japan and is covered by Japanese public insurance.

Outcome | Innovating Medical Testing Using AI on AWS

Having successfully completed its goal of creating a medical device that could effectively test for influenza infection, Aillis is looking to continue innovating its device. The company is experimenting with generative AI to determine future services and plans to expand the functionality of the AI device to include the diagnostic aids of other infections and lifestyle diseases. Aillis will continue to use AWS services to improve its solutions.

“By turning to AWS, we could launch the AI device,” says Fukuda. “We are truly satisfied using AWS services.”

About Aillis Inc.

Aillis Inc. is a medical company that focuses on developing, manufacturing, and distributing medical devices in Japan and internationally. It uses AI technology to develop its medical devices.

AWS Services Used

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 700 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)



PyTorch on AWS

PyTorch on AWS is an open-source deep learning (DL) framework that accelerates the process from ML research to model deployment.

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Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

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AWS Fargate

AWS Fargate is a serverless, pay-as-you-go compute engine that lets you focus on building applications without managing servers.

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